

Coordinate-Explicit and Route-Specific Finite-Precision Qualification of Tangent-Only Perturbations in Coupled Finite-Element Linearizations

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Abstract

Matrix norms and condition estimates quantify perturbation amplitude or worst-direction sensitivity, but they do not determine the response to one prescribed forcing direction or whether that response is distinguishable from finite-precision error. This study develops a coordinate-explicit, route-specific error-enclosure method for tangent-only perturbations in coupled finite-element linearizations. The method combines an exact finite-response identity, directional non-determination, coordinate covariance, and two a posteriori enclosures associated with distinct operation graphs: direct subtraction of parent solutions and solution of the response equation. Each enclosure covered 300/300 development systems and 300/300 newly generated holdout systems under both decimal-source and binary64-operand reference contracts. An independent 160-digit Decimal implementation checked 72 stratified lane comparisons with a maximum normalized difference of 6.96×10^{-153} . The response equation had lower reference error in 295/300 development and 271/300 new-holdout decimal-source cases, but no universal route ranking followed. Effectivity remained conservative. Its factorization showed median Frobenius-majorant inflation near 1.000–1.005, whereas median directional-alignment inflation ranged from 1.36×10^3 to 2.31×10^4 . Replication on two archived coupled finite-element tangents of dimensions 196 and 900 then demonstrated route-specific threshold qualification on application operators. The

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resulting procedure provides a reproducible reference benchmark for evaluating future scalable response estimators.

Keywords: tangent matrix, matrix perturbation, finite-element linearization, coordinate scaling, finite-precision error enclosure, directional response

1. Introduction

Coupled nonlinear finite-element models repeatedly form block tangents whose unknowns carry different units and characteristic scales. Implementations may modify the matrix passed to a linear solver without adding the same term to the residual. We call this a *tangent-only matrix perturbation*. Such a perturbation is distinct from residual-consistent stabilization and from a consistent algorithmic tangent, whose derivative must match the discrete constitutive update [1]. It is also distinct from inexact-Newton forcing, pseudo-transient continuation, conditioning-oriented finite-element reformulation, and preconditioning, which target nonlinear globalization, operator conditioning, or linear-solver performance rather than the response to one prescribed tangent modification [2, 3, 4, 5, 6, 7, 8].

Classical diagnostics do not answer the complete selected-response question. A relative matrix norm measures perturbation amplitude in one representation; a condition estimate bounds worst-direction sensitivity; structured or componentwise conditioning describes admissible perturbation patterns; scaling changes the numerical coordinates; and verified or mixed-precision methods assess a prescribed solve [9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28]. These tools remain indispensable, but neither matrix amplitude nor a worst-direction bound determines the realized action of ΔJz . A second distinction then appears in finite precision: the same exact response can be evaluated by subtracting two parent solutions or by solving its response equation, and the two operation graphs propagate different errors.

The computational advance developed here is the joint treatment of those two missing objects. The method first preserves the coordinate representation and selected forcing direction, then derives a posteriori finite-precision error enclosures for both operation graphs. The direct route propagates two parent-solve errors and subtraction error. The response route propagates the unperturbed-parent error, forcing-construction defect, and response-solve residual through the perturbed inverse. Consequently, route-specific intervals can be compared with a threshold

without treating a matrix norm, a condition number, or raw route error as a substitute for the realized response.

Four contributions follow. First, Propositions 1–3 formalize the exact finite response, prove that matched matrix and worst-direction metrics do not determine a selected response, and distinguish coordinate covariance from same-numeric redeclaration. Second, Propositions 4 and 5 derive direct-difference and response-equation error enclosures, and Corollary 1 gives a sound threshold relation. Third, both enclosures are evaluated on a common 300-system development design and a newly generated 300-system holdout under decimal-source and binary64-operand reference contracts; an independent Decimal backend audits a stratified subset. Fourth, effectivity is decomposed into norm-majorant and directional-alignment inflation, then the complete procedure is replicated on two archived coupled finite-element tangents. The contribution is a reference-level response-enclosure method; sparse production estimation and controlled large-system cost validation remain a separate problem.

Table 1 identifies the object supplied beyond the closest computational contexts. Figure 1 gives the method architecture, and Algorithms 1–3 specify the two route implementations.

Table 1: Closest computational contexts and the response object supplied by this work.

Context	Primary object	Missing selected-response object	Supplied here
Normwise and structured perturbation analysis	Matrix change and worst-case sensitivity	Realized action for one b and ΔJ	Directional response identity and non-determination result
Scaling and equilibration	Coordinate representation and conditioning	Same-operator versus redeclared-operator distinction	Coordinate-covariant perturbation contract
Residual-based or verified solve analysis	Error in one prescribed linear solve	Error propagation through two response operation graphs	Route-specific a posteriori enclosures
Newton regularization and continuation	Nonlinear or linear convergence behavior	Finite-precision response of one frozen tangent	Threshold-qualified local response only

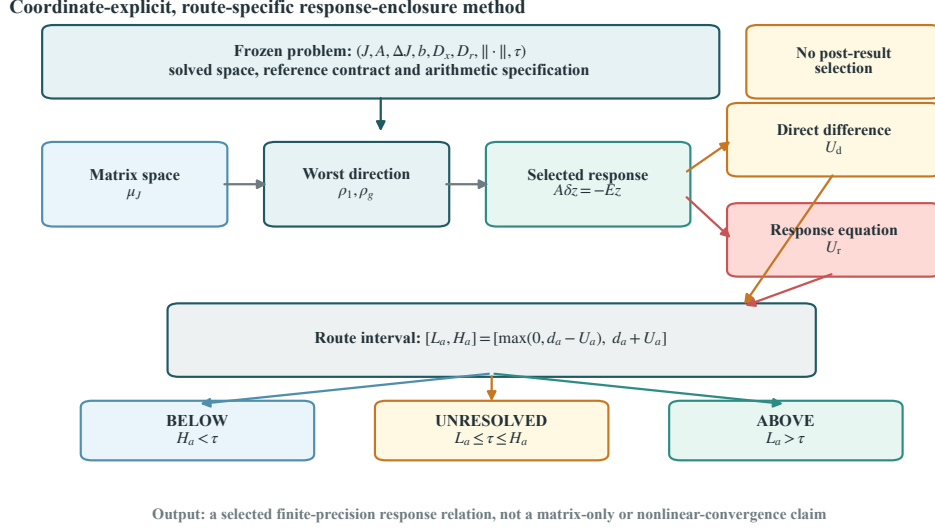


Figure 1: Coordinate-explicit response-enclosure architecture. Matrix and inverse measures precede the selected response, after which the direct-difference and response-equation operation graphs produce separate finite-precision intervals. An external threshold gives above, below, or unresolved outcomes.

Section 2 defines the response and algorithms. Sections 3 and 4 establish directional and coordinate properties. Section 5 derives and validates both error enclosures and analyzes their effectivity. Section 6 applies the method to two coupled finite-element linearizations, and Section 7 integrates the computational implications and scope.

2. Mathematical and Computational Framework

2.1. Problem definition and solved-space contract

Let J and $J + \Delta J$ be square, nonsingular operators on one solved space, with a common right-hand side b and one coordinate representation. A constrained problem enters only after its free or otherwise well-posed reduced space is specified. The vector norm and subordinate matrix norm are fixed together. The comparison therefore excludes changes in load, coordinates, nullspace, or residual root; under these assumptions the finite response identity has a single interpretation [10, 11].

2.2. Matrix, operator, and directional measures

Let z , z_g , and $\delta z = z_g - z$ denote the unperturbed solution, perturbed solution, and response. The retained quantities answer separate questions: μ_J measures matrix amplitude; ρ_1 and ρ_g measure first-order and finite worst-direction amplification; η_z measures the selected response; G_g compares that response with matrix amplitude; and $\kappa(J)$ measures normwise sensitivity. An operator norm is not a selected-probe response, and a condition number is not a mechanism classifier [9, 11]; the quantities form an assessment chain, not a composite score.

Figure 2 places a tangent-only perturbation within a generic mixed-field block operator without assigning physical meaning to one scalar diagonal coefficient.

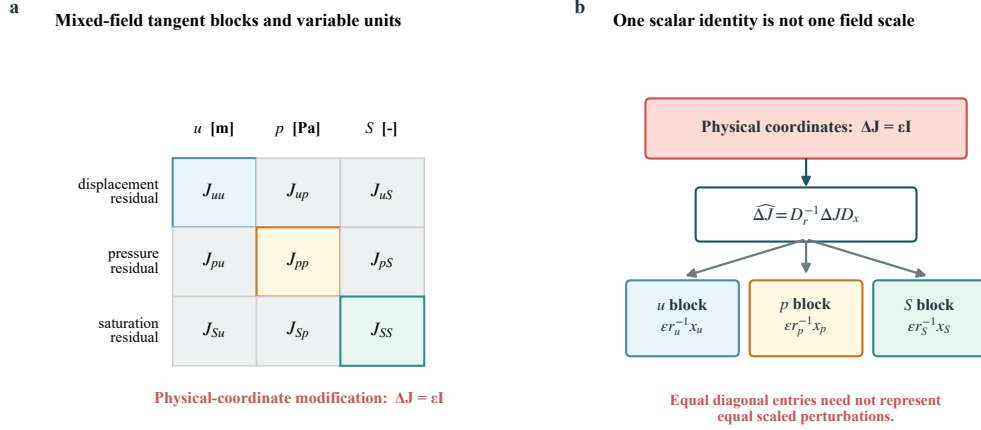


Figure 2: Mixed-field tangent anatomy. (a) Coupled displacement, pressure, and saturation blocks carry distinct variable units. (b) A scalar identity perturbation in physical coordinates becomes field-scale dependent after coordinate transformation.

Equation (1) defines the solved systems and retained metrics. Table 2 lists their roles and their non-substitution rules.

Table 2: Retained metrics and their non-substitution rules.

Metric	Question answered	Does not establish
μ_J	Matrix-space amplitude	Realized response
ρ_1	First-order worst-direction amplification	Finite selected-probe response
ρ_g	Finite worst-direction amplification	Response for one right-hand side
η_z	Realized response for one probe	Worst-case behavior
G_g	Diagnostic response/amplitude ratio	Condition number or cause
$\kappa(J)\mu_J$	Worst-case normwise comparison	Directional mechanism or trajectory

Note: Every metric requires a specified solved space, coordinates, compatible norm, right-hand side, perturbation route, and reference problem as applicable. Full declarations are listed in the Supplementary Information.

$$\begin{aligned}
Jz &= b, & (J + \Delta J)z_g &= b, & \delta z &= z_g - z, \\
\mu_J &= \frac{\|\Delta J\|}{\|J\|}, & \rho_1 &= \|J^{-1}\Delta J\|, & \rho_g &= \|(J + \Delta J)^{-1}\Delta J\|, \\
\eta_z &= \frac{\|\delta z\|}{\|z\|}, & G_g &= \frac{\eta_z}{\mu_J} \quad (\mu_J > 0), & \kappa(J) &= \|J\| \|J^{-1}\|.
\end{aligned} \tag{1}$$

2.3. Exact finite perturbation response

Proposition 1 (Exact finite response). *Let J and $A = J + \Delta J$ be nonsingular, with $Jz = b$ and $Az_g = b$. Then $\delta z = z_g - z$ satisfies*

$$\begin{aligned}
(J + \Delta J)\delta z &= -\Delta Jz, \\
\delta z &= -(J + \Delta J)^{-1}\Delta Jz.
\end{aligned} \tag{2}$$

No symmetry, definiteness, or small-perturbation assumption is required.

Proof. Substitution of $z_g = z + \delta z$ into $Az_g = b = Jz$ gives $A\delta z = -\Delta Jz$. Nonsingularity of A yields the second identity. $\square$

The identity is classical perturbation algebra [10, 11]; its role here is to expose

82 the realized forcing direction ΔJz and the operator that propagates it. It is not itself
 83 a floating-point accuracy statement.

84 Equation (3) follows from the subordinate norm and gives $\eta_z \leq \rho_g$ for nonzero
 85 z . This is a worst-direction bound, whereas η_z evaluates only the direction gen-
 86 erated by the selected solution and perturbation. Equality may occur under align-
 87 ment but is not implied for an arbitrary right-hand side. The bound remains finite
 88 whenever the perturbed operator is nonsingular, even when the Neumann condi-
 89 tion below is unavailable.

$$90 \quad \eta_z \leq \rho_g. \quad (3)$$

91 Equation (4) uses $E = J^{-1}\Delta J$, for which the response operator is $(I + E)^{-1}E$.
 92 If $\rho_1 = \|E\| < 1$, the Neumann series bounds the response by $\rho_1/(1 - \rho_1)$. The
 93 denominator is essential; ρ_1 alone is only a first-order interpretation as $\rho_1 \rightarrow 0$.
 94 For $\rho_1 \geq 1$, Proposition 1 still holds for nonsingular matrices, but this Neumann
 95 argument provides no bound [10, 11].

$$96 \quad \rho_1 < 1, \quad \eta_z \leq \frac{\rho_1}{1 - \rho_1}. \quad (4)$$

97 Equation (5) records the submultiplicative estimate $\rho_1 \leq \kappa(J)\mu_J$ in the same
 98 norm. If this product is below one, substitution in the Neumann result gives
 99 the stated finite comparison with its denominator. The estimate is coordinate-
 100 dependent, normwise, and worst case; it contains no information about the direc-
 101 tion ΔJz . It can bound sensitivity under its assumptions but cannot identify the
 102 mechanism of one observed response [9, 11].

$$103 \quad \begin{aligned} &\rho_1 \leq \kappa(J)\mu_J, \\ &\kappa(J)\mu_J < 1 \implies \eta_z \leq \frac{\kappa(J)\mu_J}{1 - \kappa(J)\mu_J}. \end{aligned} \quad (5)$$

104 Changing b adds a data-perturbation term, and changing the residual or its root
 105 is not a tangent-only comparison. A singular unreduced operator first requires a
 106 well-posed quotient, projection, generalized inverse, or reduced formulation [29,
 107 30]. Finally, the identities compare two local linear solves at one state; Newton
 108 acceptance, globalization, basin, and final root require separate nonlinear evidence
 109 [2, 31].

110 *2.4. Computational assessment procedure*

111 Algorithms 1–3 turn the mathematical contracts into two explicit computa-
 112 tional graphs. The declaration stage freezes the solved space, coordinates, ref-
 113 erence contract, and arithmetic specification. The direct graph propagates two
 114 parent-solve errors and subtraction error; the response graph propagates parent,
 115 forcing-construction, and response-solve errors. Section 4 establishes the transfor-
 116 mation requirement, and Section 5 proves the corresponding enclosures.

Algorithm 1 Coordinate and reference-problem declaration

Require: J , guarded operand A , declared perturbation ΔJ , b , coordinate maps D_x, D_r , norm $\|\cdot\|$, computational route, reference contract, arithmetic specification, and optional threshold τ

Ensure: A frozen solved-space problem and the realized perturbation $E = A - J$

- 1: Restrict the system to the prescribed well-posed solved space
 - 2: Transform J , A , ΔJ , and b consistently with D_x, D_r
 - 3: Set $E \leftarrow A - J$ in the selected reference contract
 - 4: Check dimensions, finiteness, operand identity, and reference-contract separation
 - 5: Freeze the route, precision, norm, threshold, and output fields before evaluation
-

Algorithm 2 Direct-difference finite-precision error enclosure

Require: Frozen output of Algorithm 1

Ensure: $\tilde{\delta}_d, U_d, [L_d, H_d]$

- 1: Compute $\tilde{z} \approx J^{-1}b$ and $\tilde{z}_g \approx A^{-1}b$
 - 2: Form $r_z = b - J\tilde{z}$ and $r_{z_g} = b - A\tilde{z}_g$
 - 3: Obtain majorants $\beta_J \geq \|J^{-1}\|_2$ and $\beta_A \geq \|A^{-1}\|_2$
 - 4: Set $U_z \leftarrow \beta_J \|r_z\|_2$ and $U_{z_g} \leftarrow \beta_A \|r_{z_g}\|_2$
 - 5: Compute $\tilde{\delta}_d \leftarrow \text{fl}(\tilde{z}_g - \tilde{z})$ and its subtraction bound U_{sub}
 - 6: Set $U_d \leftarrow U_z + U_{z_g} + U_{\text{sub}}$
 - 7: Set $d_d \leftarrow \|\tilde{\delta}_d\|_2$, $L_d \leftarrow \max(0, d_d - U_d)$, and $H_d \leftarrow d_d + U_d$
 - 8: Report the relation of $[L_d, H_d]$ to τ
-

Algorithm 3 Response-equation finite-precision error enclosure

Require: Frozen output of Algorithm 1

Ensure: $\tilde{\delta}_r$, U_r , $[L_r, H_r]$

- 1: Compute $\tilde{z} \approx J^{-1}b$ and $r_z = b - J\tilde{z}$
 - 2: Obtain $\beta_J \geq \|J^{-1}\|_2$, $\beta_A \geq \|A^{-1}\|_2$, and $\alpha_E \geq \|E\|_2$; set $U_z \leftarrow \beta_J \|r_z\|_2$
 - 3: Construct the route operand $\tilde{f} \leftarrow \text{fl}(-\Delta J\tilde{z})$
 - 4: Compute $\tilde{\delta}_r \approx A^{-1}\tilde{f}$
 - 5: Form $q_f = \tilde{f} + E\tilde{z}$ and $r_\delta = \tilde{f} - A\tilde{\delta}_r$
 - 6: Set $U_{\text{parent}} \leftarrow \beta_A \alpha_E U_z$, $U_{\text{force}} \leftarrow \beta_A \|q_f\|_2$, and $U_{\text{solve}} \leftarrow \beta_A \|r_\delta\|_2$
 - 7: Set $U_r \leftarrow U_{\text{parent}} + U_{\text{force}} + U_{\text{solve}}$
 - 8: Set $d_r \leftarrow \|\tilde{\delta}_r\|_2$, $L_r \leftarrow \max(0, d_r - U_r)$, and $H_r \leftarrow d_r + U_r$
 - 9: Report the relation of $[L_r, H_r]$ to τ
-

117 The architecture is not an arbitrary checklist. Table 3 links each component to
118 a specific omission exposed by the existing evidence. The conclusions are inten-
119 tionally local: they establish what is needed to interpret one selected response, not
120 prevalence across operator populations.

Table 3: Evidence-backed necessity of the assessment components.

Component	Frozen evidence	Bounded conclusion	Not inferred
Directional response	P2 matches μ_J, ρ_1, ρ_g , and $\kappa(J)$ while η_z differs	One selected response needs directional information	Frequency or population law
Coordinate-consistent mapping	P3 transformation versus same-label redeclaration	One numeric label need not denote one operator	Preferred scaling or perturbation
Reference-problem separation	Historical 287/300 versus revised separate reference problems	A mismatched reference can invalidate enclosure use	Perturbation defect or cause
Route-specific uncertainty	Direct and response-equation operation graphs	One uncertainty budget cannot be transferred automatically	Universal route ranking

121 3. Analytical Verification

122 **Proposition 2** (Directional non-determination). *For every $\varepsilon > 0$, there exist two*
 123 *perturbations with identical μ_J , ρ_1 , ρ_g , and $\kappa_2(J)$ but different realized responses*
 124 *η_z . Hence those scalar matrix and worst-direction quantities do not determine the*
 125 *response for a selected right-hand side.*

126 *Construction and proof.* Fix the Euclidean vector norm and spectral matrix norm,
 127 take $J = I_2$, and choose the first coordinate as the right-hand side. Two rank-
 128 one diagonal perturbations with the same amplitude ε act respectively along and
 129 orthogonally to the unperturbed solution. Equation (6) isolates direction while
 130 holding the operator, norm, right-hand side, perturbation amplitude, and condition
 131 number fixed.

$$132 \quad \begin{aligned} J &= I_2, & b &= e_1 = [1, 0]^T, \\ \Delta J_A &= \varepsilon \operatorname{diag}(1, 0), & \Delta J_B &= \varepsilon \operatorname{diag}(0, 1), \quad \varepsilon > 0. \end{aligned} \quad (6)$$

133 The matrix and operator metrics are matched in the two cases. Since $\|J\| = 1$,
 134 both perturbations give $\mu_J = \varepsilon$; since $J^{-1} = I_2$, both give $\rho_1 = \varepsilon$; and since the
 135 guarded inverse scales only the perturbed coordinate, both give $\rho_g = \varepsilon/(1 + \varepsilon)$.
 136 The spectral condition number is one in each case. Thus even the combination
 137 of matrix amplitude, unguarded inverse amplification, guarded inverse amplifica-
 138 tion, and condition number is identical. This equality is stronger than matching
 139 only $\|\Delta J\|$: the two perturbations also have the same finite worst-direction bound.
 140 What remains unspecified by those metrics is whether the selected solution lies in
 141 the range of the perturbation. The construction therefore prepares a direct test of
 142 whether matched matrix-level and operator-level quantities determine one realized
 143 response.

$$144 \quad \begin{aligned} \mu_J(A) &= \mu_J(B) = \varepsilon, \\ \rho_1(A) &= \rho_1(B) = \varepsilon, \\ \rho_g(A) &= \rho_g(B) = \frac{\varepsilon}{1 + \varepsilon}, \\ \kappa_2(J) &= 1. \end{aligned} \quad (7)$$

145 Equation (7) shows that the constructions have identical matrix amplitude,
 146 inverse-mediated measures, and condition number. Their realized responses nev-
 147 ertheless separate. In case A, the perturbation acts on $z = e_1$, giving $\eta_z = \varepsilon/(1 + \varepsilon)$

148 and attaining the guarded-inverse bound. In case B, $\Delta J_B z = 0$, so the guarded and
 149 unguarded solutions coincide and $\eta_z = 0$. The missing information is directional
 150 alignment between the solution, perturbation range, and propagated response. This
 151 proves Proposition 2.

152 Figure 3 contrasts the aligned and orthogonal constructions under matched ma-
 153 trix and worst-direction quantities.

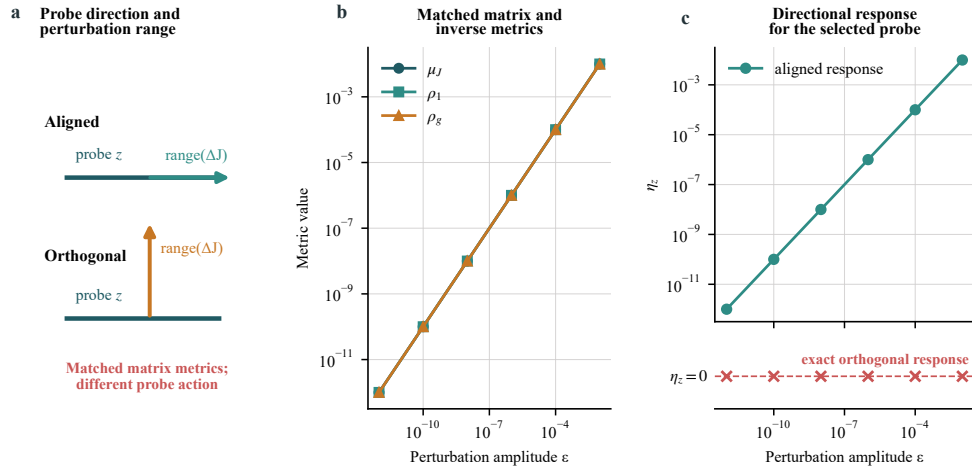


Figure 3: Directional counterexample. Alignment geometry separates the selected probe from the perturbation range; matrix and worst-direction quantities remain matched while the realized response differs for six pre-specified perturbation amplitudes.

154
$$\eta_z(A) = \frac{\epsilon}{1 + \epsilon}, \quad \eta_z(B) = 0. \quad (8)$$

155 Equation (8) records the exact directional separation used in the numerical
 156 reproduction.

157 The pre-specified numerical reproduction evaluated six frozen amplitudes
 158 from 1×10^{-12} to 1×10^{-2} using an independent high-precision route for the
 159 exact construction. Across all six rows, the maximum relative difference between
 160 the paired matrix metrics was 0, the maximum relative error in the nonzero
 161 aligned response was $8.721173782511926 \times 10^{-102}$, and the maximum absolute
 162 orthogonal response was 0. The implementation check summarized in Eq. (9)
 163 reproduces the analytical metric match and the response distinction without

164 changing the construction after observation. The numerical verification validates
 165 implementation of this exact example; it does not add an empirical frequency
 166 statement. In particular, six amplitudes for one two-dimensional construction are
 167 not an operator-family survey, and the zero response in case B follows from exact
 168 orthogonality rather than from a general perturbation-screening rule.

$$\begin{aligned}
 & \max \delta_{\text{matched}} = 0, \\
 169 \quad & \max \delta_{\eta_A} = 8.721173782511926 \times 10^{-102}, \\
 & \max |\eta_B| = 0.
 \end{aligned} \tag{9}$$

170 Proposition 2 is an existence result rather than a frequency statement. The
 171 numerical reproduction verifies the implementation of its exact construction; the
 172 synthetic-family and finite-element evidence below address separately designed
 173 operator sets.

174 4. Coordinate-Transformation Verification

175 **Proposition 3** (Coordinate covariance and redeclaration non-equivalence). *Under*
 176 *$z = D_x \hat{z}$ and $\hat{r} = D_r^{-1} r$, one fixed perturbed operator transforms as $\hat{J} = D_r^{-1} J D_x$*
 177 *and $\widehat{\Delta J} = D_r^{-1} \Delta J D_x$. A same-numeric perturbation redeclared after the coordi-*
 178 *nate change is generally a different physical operator.*

179 Let the physical unknown and scaled unknown be related by the nonsingular
 180 matrix D_x , and let residual coordinates be transformed by the nonsingular matrix
 181 D_r . The frozen convention is $z = D_x \hat{z}$ and $\hat{r} = D_r^{-1} r$. Substitution into $r = Jz - b$
 182 defines the scaled operator and right-hand side as $\hat{J} = D_r^{-1} J D_x$ and $\hat{b} = D_r^{-1} b$. A
 183 perturbation belongs to the same operator relation and must therefore transform
 184 as $\widehat{\Delta J} = D_r^{-1} \Delta J D_x$. Equation (10) defines equivalent coordinates only when the
 185 unknown, residual, right-hand side, operator, and perturbation are transformed
 186 together. Applying a transformation to some objects while independently redefin-
 187 ing another object creates a different perturbed problem, even if the same numeric
 188 symbol is displayed. Scaling and equilibration literature motivates explicit coordi-
 189 nate declarations but does not supply unique physical choices for D_x and D_r
 190 [12, 13, 14, 15].

$$\begin{aligned}
z &= D_x \hat{z}, & \hat{r} &= D_r^{-1} r, \\
\hat{f} &= D_r^{-1} J D_x, & \hat{b} &= D_r^{-1} b, \\
\widehat{\Delta J} &= D_r^{-1} \Delta J D_x.
\end{aligned} \tag{10}$$

Suppose first that the perturbation is declared in physical coordinates as $\Delta J_{\text{phys}} = \varepsilon I$. Equation (11) gives its representation in the scaled operator is obtained by the joint transformation, giving $\widehat{\Delta J} = \varepsilon D_r^{-1} D_x$. The transformed perturbation is a scalar identity only when $D_r^{-1} D_x$ is a scalar multiple of the identity, and it retains the same εI representation only when $D_x = D_r$. In a mixed-dimensional system, different unknown and residual scales generally make the diagonal entries field dependent. This is a coordinate statement: it identifies the scaled representation of one fixed physical operator. It neither changes that operator nor evaluates the practical consequences of its action. Treating the transformed matrix as if it had to retain the original scalar identity form would replace transformation by redeclaration.

$$\Delta J_{\text{phys}} = \varepsilon I, \quad \widehat{\Delta J} = \varepsilon D_r^{-1} D_x. \tag{11}$$

The reverse construction begins with a perturbation declared in scaled coordinates as $\widehat{\Delta J} = \beta I$. Equation (12) maps it to physical coordinates and gives $\Delta J_{\text{phys}} = \beta D_r D_x^{-1}$, which is generally a field-dependent diagonal operator. Thus a dimensionless identity guard denotes a family of physical perturbations indexed by the declared unknown and residual scales. The mapping does not select those scales; it only states which physical operator corresponds to the scaled identity after the coordinate contract is fixed. Consequently, the label βI is incomplete without the operator space, D_x , D_r , and the direction in which the transformed solution is interpreted. This distinction also prevents a same-numeric identity written in two coordinate systems from being treated as one perturbation without checking the transformation rule.

Figure 4 distinguishes equivalent coordinate transformation from same-numeric-label redeclaration.

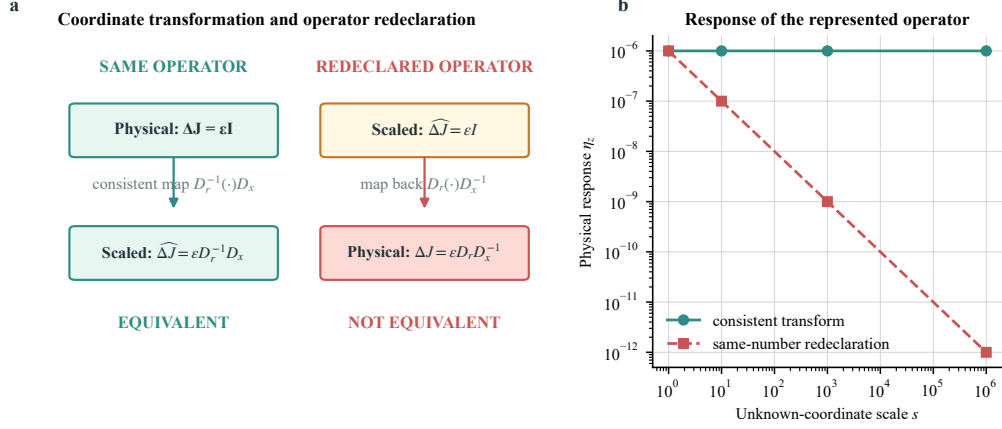


Figure 4: Coordinate transformation versus operator redeclaration. Consistent transformation preserves the represented perturbation, whereas reusing a same-numeric identity in scaled coordinates maps back to a different physical perturbation.

$$\widehat{\Delta J} = \beta I, \quad \Delta J_{\text{phys}} = \beta D_r D_x^{-1}. \quad (12)$$

The equivalent-transformation calculation used a two-dimensional construction with pre-specified scale factors 1, 10, 1×10^3 , and 1×10^6 . In the equivalent transformed-operator route, the unknown, right-hand side, operator, and perturbation were transformed consistently. The resulting maximum relative difference between physical-equivalent perturbation operators was $7.490674676829096 \times 10^{-102}$, and the relative spread of the physical response was 0. The check in Eq. (13) validates the implemented transformation for the frozen construction and pre-specified scales. They do not establish that arbitrary independently specified guards are invariant; invariance here follows because the transformed descriptions represent the same physical operator and the same physical solution response. The numerical values therefore check the algebraic mapping rather than define a coordinate-independent scalar guard.

$$\begin{aligned} \max \delta_{\text{operator}} &= 7.490674676829096 \times 10^{-102}, \\ \text{spread}(\eta_z)_{\text{transformed}} &= 0. \end{aligned} \quad (13)$$

The second pre-specified lane reapplied the same numeric identity after changing coordinates instead of transforming the original perturbation. With $D_r = I_2$

233 and $D_x = \text{diag}(1, s)$, that scaled identity maps back to a physical perturbation
 234 whose second diagonal entry is ε/s , whereas the original physical identity retains ε
 235 on both diagonal entries. The two physical perturbations coincide at $s = 1$ and dif-
 236 fer for every pre-specified $s > 1$; the pre-specified inequality check was therefore
 237 true. Equation (14) demonstrates that a shared numeric label does not establish op-
 238 erator equivalence. It does not show that one of the resulting operators should be
 239 selected. The observed response change follows from comparing different physi-
 240 cal perturbations, not from a failure of coordinate transformation.

$$241 \quad \Delta J_{\text{redeclared}} \neq \Delta J_{\text{transformed}} \quad \text{for every pre-specified } s > 1. \quad (14)$$

242 Proposition 3 is a covariance requirement, not a scale-selection rule. Equilibra-
 243 tion and physical nondimensionalization may pursue different objectives [14, 15];
 244 the present result requires only that an operator comparison preserve its declared
 245 coordinates or explicitly acknowledge a redeclared perturbation.

246 5. Finite-Precision Error-Enclosure Method and Validation

247 5.1. Two computational routes

248 Let $A = J + E$ denote the guarded operand in one fixed reference contract. The
 249 exact parent and response problems are

$$250 \quad Jz = b, \quad Az_g = b, \quad A\delta z = -Ez, \quad \delta z = z_g - z. \quad (15)$$

251 Here $E = A - J$ is the realized perturbation of the two solve operands. The
 252 declared ΔJ remains the operand used by the floating-point forcing construction.
 253 This distinction is immaterial when addition is exact, but retains last-bit operand
 254 effects when A and J are captured independently under IEEE 754 binary arithmetic
 255 [32, 33].

256 The direct route computes two parent solutions and subtracts them,

$$257 \quad \tilde{\delta}_d = \text{fl}(\tilde{z}_g - \tilde{z}) = \delta z + e_g - e + q_{\text{sub}}, \quad (16)$$

258 where $e = \tilde{z} - z$, $e_g = \tilde{z}_g - z_g$, and q_{sub} is the subtraction error.

259 **Proposition 4** (Direct-difference error enclosure). *If $\|e\|_2 \leq U_z$, $\|e_g\|_2 \leq U_{z_g}$, and*

260 $\|q_{\text{sub}}\|_2 \leq U_{\text{sub}}$, then

$$261 \quad \|\tilde{\delta}_d - \delta z\|_2 \leq U_d := U_z + U_{z_g} + U_{\text{sub}}. \quad (17)$$

262 *Proof.* Subtract δz from Eq. (16) and apply the triangle inequality. $\square$

263 The response route constructs $\tilde{f} = \text{fl}(-\Delta J \tilde{z})$ and solves $A \tilde{\delta}_r \approx \tilde{f}$. Define the
264 forcing defect $q_f = \tilde{f} + E \tilde{z}$ and response residual $r_\delta = \tilde{f} - A \tilde{\delta}_r$.

265 **Proposition 5** (Response-equation error enclosure). Assume $\|\tilde{z} - z\|_2 \leq U_z$ and
266 let $\beta_A \geq \|A^{-1}\|_2$ and $\alpha_E \geq \|E\|_2$. Then

$$267 \quad \begin{aligned} \|\tilde{\delta}_r - \delta z\|_2 &\leq U_r, \\ U_r &:= \underbrace{\beta_A \alpha_E U_z}_{U_{\text{parent}}} + \underbrace{\beta_A \|q_f\|_2}_{U_{\text{force}}} + \underbrace{\beta_A \|r_\delta\|_2}_{U_{\text{solve}}}. \end{aligned} \quad (18)$$

268 *Proof.* Because $A \delta z = -E z$ and $A \tilde{\delta}_r = \tilde{f} - r_\delta$,

$$A(\tilde{\delta}_r - \delta z) = q_f - E(\tilde{z} - z) - r_\delta.$$

269 Multiplication by A^{-1} , submultiplicativity, and the assumed majorants give
270 Eq. (18). $\square$

271 Propositions 4 and 5 enclose the same exact response through different opera-
272 tion graphs. Their components therefore cannot be exchanged merely because the
273 observed response vectors are close.

274 5.2. Parent bounds, threshold relation, and reference contracts

275 For a computed solution $\tilde{x}$ of $Bx = c$, residual $r = c - B\tilde{x}$, and any $\beta_B \geq$
276 $\|B^{-1}\|_2$, the parent error satisfies

$$277 \quad \|\tilde{x} - x\|_2 \leq \|B^{-1}\|_2 \|r\|_2 \leq \beta_B \|r\|_2. \quad (19)$$

278 The implementation uses

$$279 \quad \beta_B = \|B^{-1}\|_F \geq \|B^{-1}\|_2, \quad \alpha_E = \|E\|_F \geq \|E\|_2. \quad (20)$$

280 Thus the Frobenius quantities are explicit majorants; they are not subordinate ma-
281 trix norms paired with the Euclidean vector norm. For binary64 subtraction, the

282 adopted allowance is

$$283 \quad U_{\text{sub}} = \gamma_1 \| \tilde{z}_g \| + \| \tilde{z} \|_2 + \sqrt{n} 2^{-1075}, \quad \gamma_1 = \frac{2^{-53}}{1 - 2^{-53}}. \quad (21)$$

284 For either route $a \in \{d, r\}$, let $d_a = \| \tilde{\delta}_a \|_2$ and let U_a be the matching enclosure.
 285 The response-magnitude interval is

$$286 \quad L_a = \max(0, d_a - U_a), \quad H_a = d_a + U_a. \quad (22)$$

287 **Corollary 1** (Sound threshold relation). *If Proposition 4 or 5 applies, then $L_a \leq$*
 288 *$\| \delta z \|_2 \leq H_a$. Hence, for a threshold $\tau \geq 0$ fixed independently of the observed*
 289 *response, $L_a > \tau$ proves that the exact response exceeds τ , and $H_a < \tau$ proves*
 290 *that it lies below τ .*

291 **Definition 1** (Route-qualified threshold relation). The outcome is *above* if $L_a > \tau$,
 292 *below* if $H_a < \tau$, and *unresolved* otherwise. Setting $\tau = 0$ recovers zero exclusion
 293 as a special case.

294 Two reference contracts are retained. The decimal-source contract treats the
 295 registered decimal matrices and vectors as the real-valued problem. The binary64-
 296 operand contract reconstructs the captured IEEE operands exactly and treats those
 297 values as a separate real-valued problem. Each residual, inverse majorant, forc-
 298 ing defect, and reference response is evaluated only within its matching contract.
 299 Figure 5 shows both reference lanes and both operation graphs.

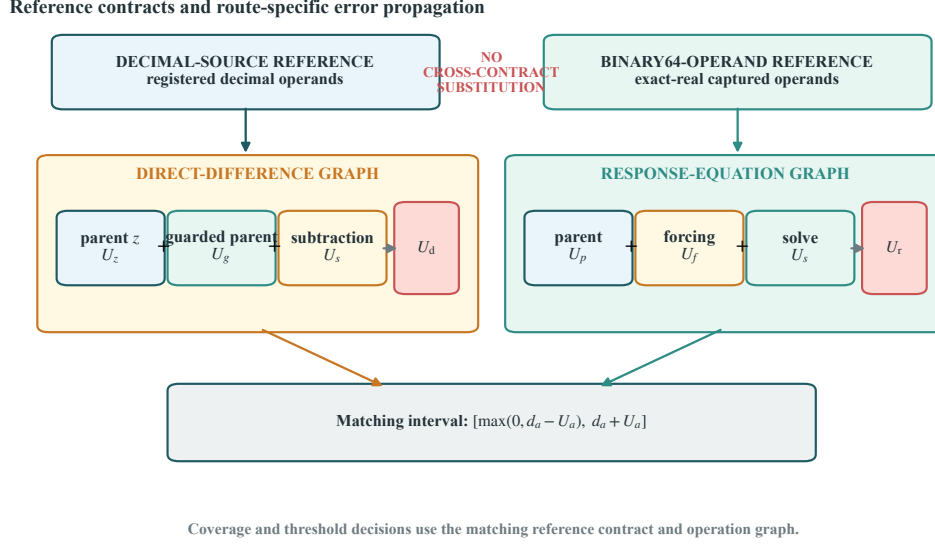


Figure 5: Reference-contract and operation-graph anatomy. Decimal-source and binary64-operand problems remain separate. Within either contract, direct subtraction combines two parent and subtraction components, whereas the response equation combines parent propagation, forcing construction, and response-solve components.

300 5.3. Prospective enclosure validation

301 The direct-route model was first evaluated on a 300-system development set
 302 and a disjoint 300-system holdout containing SPD, nonsymmetric, and saddle-
 303 point-like matrices of dimensions 2, 4, 8, and 16. Each reference contract achieved
 304 300/300 coverage on both sets. A subsequent preregistered response-route study
 305 froze Proposition 5 and then generated a new 300-system holdout with disjoint
 306 seeds, condition exponents, and perturbation amplitudes. Both routes were evalu-
 307 ated without changing the matrices or gates.

308 Table 4 reports the result. Direct and response enclosures each covered
 309 300/300 errors in all four data-set–contract combinations. The development
 310 decimal-source response error was lower in 295/300 cases, reproducing the earlier
 311 route comparison; the corresponding count was 271/300 on the new holdout. In
 312 the binary64-operand contract, the counts were 199/300 and 160/300. Lower
 313 response error therefore did not imply a universal route ordering.

314 The 120- and 180-digit core evaluations agreed to maximum normalized dif-

Table 4: Prospective direct- and response-route enclosure validation. Coverage is shown as covered/total; the last two columns count cases in which the response route produced the smaller quantity.

Set	Reference contract	Direct coverage	Response coverage	Lower response error	Lower response enclosure
Development	Decimal source	300/300	300/300	295/300	283/300
Development	Binary64 operand	300/300	300/300	199/300	229/300
New holdout	Decimal source	300/300	300/300	271/300	229/300
New holdout	Binary64 operand	300/300	300/300	160/300	188/300

ferences of 5.16×10^{-110} on development and 6.82×10^{-107} on the new holdout, with no coverage-decision mismatch. A separately implemented 160-digit Python Decimal solver, using manual elimination and importing neither NumPy, SciPy, nor mpmath, checked 36 stratified holdout systems in both contracts. Its 72 lane comparisons agreed with maximum normalized difference 6.96×10^{-153} . The first execution and two tooling-only comparator corrections are retained in the Supplement rather than overwritten.

5.4. Effectivity anatomy

Coverage alone does not measure sharpness. For positive true route error $e_a = \|\tilde{\delta}_a - \delta z\|_2$, define $I_a = U_a/e_a$. For a parent solve whose bound is $U = \|B^{-1}\|_F \|r\|_2$ and whose true error is $e = \|B^{-1}r\|_2$, the effectivity factorizes exactly as

$$I_{\text{parent}} = \underbrace{\frac{\|B^{-1}\|_F}{\|B^{-1}\|_2}}_{\text{norm-majorant inflation}} \underbrace{\frac{\|B^{-1}\|_2 \|r\|_2}{\|B^{-1}r\|_2}}_{\text{directional-alignment inflation}}. \quad (23)$$

This decomposition connects enclosure conservativeness to the same worst-direction versus realized-direction distinction established by Proposition 2.

Across 2,400 parent records, the median norm-majorant factor was 1.0001 on development and 1.0050 on the original holdout. Median directional inflation instead ranged from 1.18×10^4 to 2.31×10^4 on development and from 1.36×10^3 to

332 1.88×10^3 on holdout (Table 5). The central conservativeness therefore came pri-
 333 marily from replacing the realized residual direction by a worst-direction inverse
 334 norm, not from using the Frobenius norm instead of the spectral norm.

Table 5: Factorization of parent-bound effectivity. Ranges span both parent systems and both reference contracts.

Set	Quantity	Median range	95th-percentile range
Development	Frobenius majorization	1.0001–1.0001	1.94–1.94
Development	Directional inflation	1.18×10^4 – 2.31×10^4	3.18×10^8 – 8.53×10^8
Original holdout	Frobenius majorization	1.0050–1.0050	2.69–2.69
Original holdout	Directional inflation	1.36×10^3 – 1.88×10^3	3.59×10^7 – 8.72×10^7

335 A main-effects model for $\log_{10} I_{\text{eff}}$ used condition exponent, $\log_{10} \epsilon$, $\log_2 n$,
 336 matrix family, and reference lane. The development fit gave adjusted $R^2 = 0.869$
 337 and drop-one partial $R^2 = 0.855$ for condition exponent. The original holdout
 338 fit gave adjusted $R^2 = 0.057$ and condition partial $R^2 = 0.029$. Conditioning was
 339 therefore dominant in the development design but did not transfer as a universal im-
 340 portance law. The largest finite direct-route ratio is reported as $\log_{10} I_{\text{eff}} = 86.90$;
 341 it arises from a 4.9995×10^{-102} reference error rather than from undercoverage.
 342 The exact value and case identifier remain in the machine-readable record.

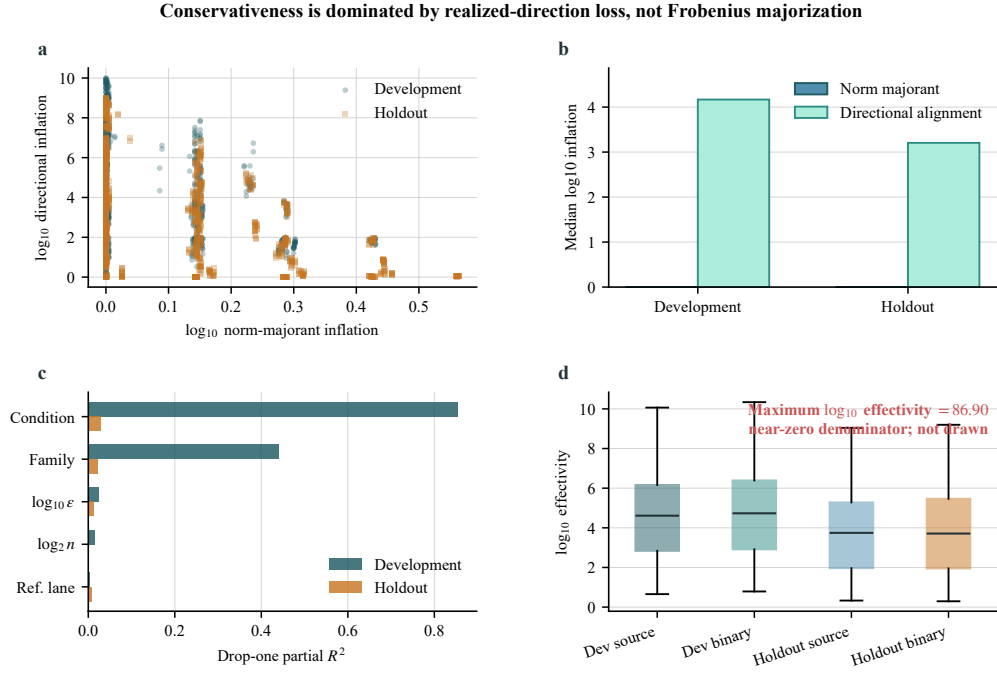


Figure 6: Effectivity anatomy and finite-grid attribution. (a) Norm-majorant inflation remains near unity while directional inflation spans orders of magnitude. (b) Median factor ranges for development and holdout. (c) Drop-one partial R^2 values from registered main-effects models. (d) Direct-route effectivity versus condition exponent, including the near-zero-denominator extreme. Partial contributions are non-additive.

343 5.5. Route comparison after enclosure validation

344 Figure 7 joins raw arithmetic behavior to valid enclosure consequences. In the
 345 decimal-source contract, the median response-to-direct error ratio was 5.29×10^{-8}
 346 on development and 2.01×10^{-3} on the new holdout; the median bound ratio
 347 was 6.90×10^{-8} and 4.11×10^{-3} , respectively. In the binary64-operand contract,
 348 median error ratios were 0.631 and 0.980, with median bound ratios 0.462 and
 349 0.480. The response graph thus reduced both error and enclosure in many cases,
 350 but its effectivity remained conservative and some cases favored the direct route.

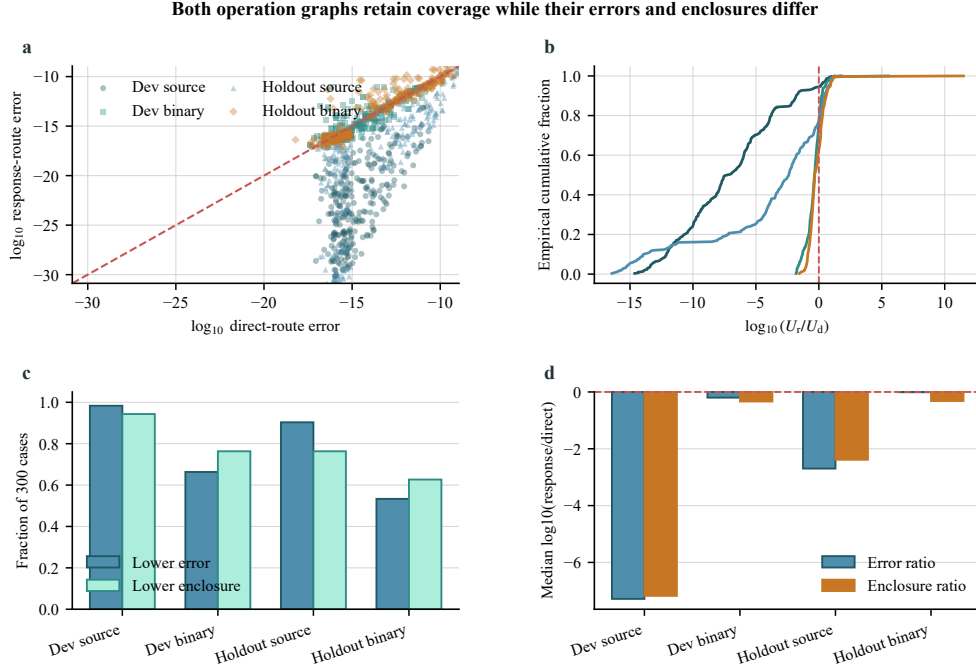


Figure 7: Validated route comparison. (a) Response- versus direct-route reference error. (b) Distribution of U_r/U_d . (c) Cases with lower response-route error and enclosure. (d) Median error and enclosure ratios by data set and reference contract. Both routes retain complete coverage; the figure does not assert universal superiority.

351 The validation establishes conditional finite-precision enclosures for the regis-
 352 tered finite grids. It does not make them interval-arithmetic certificates, calibrated
 353 probabilistic intervals, or scalable sparse estimators. Those distinctions are con-
 354 solidated here rather than repeated after each positive result.

355 6. Applications to Two Coupled Finite-Element Linearizations

356 6.1. Archived operator snapshots and reference construction

357 The application stage uses two independently archived transformed free-
 358 system snapshots from a four-field displacement–pressure–saturation discretiza-
 359 tion. M8 contains 324 total nodal degrees of freedom and 196 free unknowns;
 360 M16 contains 1156 and 900, respectively. The free ordering retains 49 and 225
 361 coordinates for each of u_x , u_y , p , and S . Unknown scales are 10^{-5} for both

displacement components, 10^7 for pressure, and 10^{-1} for saturation. Table 6 records the corresponding matrix sizes and sparsity.

Table 6: Provenance and coordinate content of the archived finite-element tangents. Each per-field count applies to u_x , u_y , p , and S .

Case	Total / free DOFs	Free DOFs per field	Stored nonzeros	Unknown-coordinate scales
M8	324 / 196	49	5,442	$u : 10^{-5}, p : 10^7, S : 10^{-1}$
M16	1,156 / 900	225	28,259	$u : 10^{-5}, p : 10^7, S : 10^{-1}$

Each snapshot fixes the IEEE binary64 matrices J and A , common right-hand side b , free-degree-of-freedom ordering, unknown and residual scales, and Euclidean comparison norm before either route is evaluated. The realized perturbation $E = A - J$ is diagonal in both cases. M8 and M16 are independently captured states rather than a controlled mesh or condition sequence.

For each case, the reference problem is the real-valued system represented exactly by the captured binary64 operands. Integer-ratio reconstruction, signed-zero checks, and the elementwise identity $A = J + E$ establish that operand contract. Parent systems and the response equation are solved independently at 120 and 180 decimal digits. Each observed response is compared only with the error enclosure for its own operation graph.

6.2. Case results and offline cost

For M8, the maximum normalized 120/180-digit difference is 8.81×10^{-121} and the maximum reference relative residual is 2.55×10^{-124} . Direct subtraction gives $d_{\text{obs}} = 2.919 \times 10^{-12}$ with $U = 5.880 \times 10^{-13}$; the response equation gives $d_{\text{obs}} = 2.920 \times 10^{-12}$ with $U = 2.588 \times 10^{-22}$. Both intervals lie above the special threshold $\tau = 0$.

For M16, the maximum normalized precision difference is 3.97×10^{-121} and the maximum reference relative residual is 4.14×10^{-124} . Direct subtraction gives $d_{\text{obs}} = 2.463 \times 10^{-11}$ with $U = 3.971 \times 10^{-12}$; the response equation gives $d_{\text{obs}} = 2.464 \times 10^{-11}$ with $U = 6.930 \times 10^{-21}$. Again, both route-specific intervals lie above $\tau = 0$. Rounded route, precision, and component summaries appear in Supplementary Tables S6–S8; exact decimal strings remain in the machine-readable ledgers.

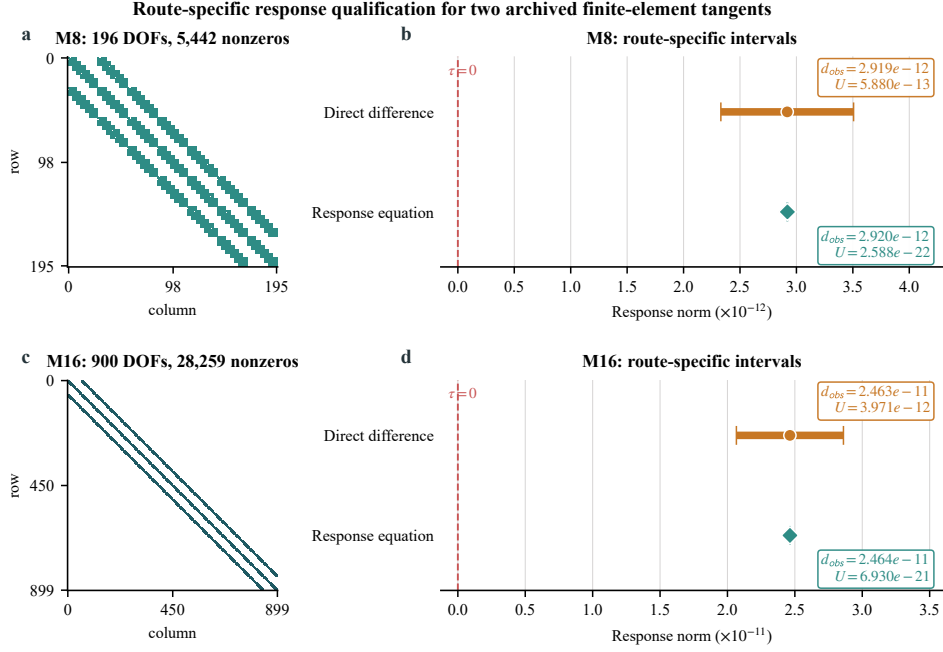


Figure 8: Route-specific response intervals for two independently archived coupled finite-element tangents. The left panels show matrix sparsity; the right panels show direct-difference and response-equation intervals on case-specific scales. Dashed references denote $\tau = 0$.

Table 7: Route-specific response intervals for the two finite-element tangents at $\tau = 0$. Displayed values are rounded; Supplementary Table S4 retains exact strings.

Case	Route	d_{obs}	U	L	U/d_{ref}
M8	Direct	2.919e-12	5.880e-13	2.330e-12	2.523e-1
M8	Response Eq.	2.920e-12	2.588e-22	2.920e-12	8.864e-11
M16	Direct	2.463e-11	3.971e-12	2.066e-11	1.922e-1
M16	Response Eq.	2.464e-11	6.930e-21	2.464e-11	2.813e-10

388 The complete M8 reference calculation, including both precision levels and an
389 independent repeat, required approximately 468 s and 110 MiB of peak memory.
390 M16 required 10.4 h and 1.35 GiB. These measurements define the implementa-
391 tion as a reproducible reference benchmark against which scalable estimators can
392 be tested; two timings do not constitute a complexity fit.

7. Discussion

7.1. The computational object beyond matrix diagnostics

The method addresses a question that matrix amplitude and conditioning leave open. The quantities μ_J , ρ_1 , ρ_g , and $\kappa(J)$ quantify representation-dependent change or worst-direction sensitivity. They do not identify whether the selected solution activates the perturbation range. Proposition 2 makes that information deficit exact: all four matrix-level quantities can be matched while the realized response changes from the guarded-inverse bound to zero. Structured and componentwise conditioning sharpen the admissible perturbation model [24, 25, 26], but a selected response still requires the direction Ez and its propagation by A^{-1} .

The second advance concerns computation of that response. Direct subtraction and the response equation are algebraically equivalent in exact arithmetic, yet they form different finite-precision operation graphs. Propositions 4 and 5 provide separate a posteriori error enclosures for those graphs. This is the central difference from a generic verified solve: the solve-error ingredients are propagated to a specific perturbation-response object, and the forcing-construction defect is retained when the declared perturbation and realized solve operands differ at the last bit. The threshold relation in Corollary 1 then turns the enclosed response into an above, below, or unresolved result without using a condition number as a surrogate decision.

7.2. Why the enclosures are conservative

Complete coverage and useful sharpness are different properties. The effectivity factorization in Eq. (23) localizes the dominant source of conservativeness. Median inflation caused by the Frobenius majorant was almost unity, whereas replacing the realized residual direction by the worst inverse direction contributed median factors between 1.36×10^3 and 2.31×10^4 . The large central ratios are therefore not primarily a Frobenius-versus-spectral artifact; they arise because a valid normwise residual bound discards directional alignment.

The multivariable analysis also prevents overinterpreting a single marginal correlation. Condition exponent explained most registered variation in the development design, but its drop-one partial R^2 fell from 0.855 to 0.029 on the original holdout. Matrix family, perturbation amplitude, and reference lane likewise changed importance across the two finite grids. The data support a mechanism—worst-direction replacement can be highly conservative—but not a universal effectivity scaling law. The extreme $\log_{10} I_{\text{eff}} = 86.90$ further illustrates why ratios

428 become unstable when the true error approaches zero even though absolute cover-
429 age remains valid.

430 7.3. *What the two routes reveal*

431 The response equation reduced decimal-source reference error in 295/300 de-
432 velopment cases and 271/300 new-holdout cases. Its enclosure was also smaller in
433 283/300 and 229/300 cases. Binary64-operand comparisons were less separated:
434 the response route had lower error in 199/300 development and 160/300 holdout
435 cases, and lower bounds in 229/300 and 188/300. These differences are consistent
436 with the operation graphs. Direct subtraction can lose small responses between
437 two complete parent solutions; the response equation avoids that subtraction but
438 inherits forcing-construction and response-solve errors. The new result is not that
439 one graph always wins, but that both can now be compared through valid route-
440 specific enclosures rather than raw arithmetic error alone.

441 7.4. *Computational-mechanics meaning and scope*

442 For a frozen coupled tangent, the method answers whether the correction in-
443 duced by a tangent-only modification is above, below, or unresolved with respect
444 to an independently supplied response threshold. A solver developer may con-
445 nect that threshold to an accepted Newton-correction tolerance, a residual scale,
446 an energy norm, or a quantity-of-interest budget. Such a choice converts a vague
447 statement that the matrix change is *small* into a testable statement about the se-
448 lected correction at one nonlinear state. M8 and M16 demonstrate this chain for
449 displacement–pressure–saturation operators with explicit field scales; both opera-
450 tion graphs excluded the special threshold zero.

451 The present implementation is intentionally a high-precision reference bench-
452 mark. It does not select a perturbation, prove nonlinear convergence, or certify
453 an engineering model. It is also not an outward-rounded interval solver. Sparse
454 estimation, controlled large-system sequences, blind transfer, and coverage–
455 tightness–cost validation belong to a separate scalable-method study. Consol-
456 idating these limits here preserves the positive result: a coordinate-consistent,
457 operation-graph-specific response can be enclosed and independently checked
458 before a production estimator is judged against it.

459 8. Conclusions

460 This study formulated tangent-only perturbation assessment as a selected finite-
461 response problem rather than a matrix-magnitude test. Exact algebra separates the
462 represented perturbation, worst-direction amplification, and realized forcing direc-
463 tion, while coordinate covariance prevents a same-numeric diagonal from being
464 mistaken for the same operator after rescaling.

465 The main computational result is a pair of a posteriori finite-precision error
466 enclosures for two algebraically equivalent but arithmetically distinct operation
467 graphs. The direct enclosure propagates two parent-solve errors and subtrac-
468 tion error; the response enclosure propagates parent error, forcing-construction
469 defect, and response-solve residual. Each route covered 300/300 development
470 and 300/300 newly generated holdout systems in both reference contracts. An
471 independent Decimal implementation confirmed a stratified 72-lane subset to
472 6.96×10^{-153} normalized agreement.

473 The analysis also explains why coverage can coexist with conservative bounds.
474 Frobenius majorization contributed median factors near unity, whereas directional-
475 alignment inflation contributed factors of 10^3 – 10^4 . The response graph often re-
476 duced both arithmetic error and enclosure size, but the binary64 and holdout results
477 ruled out a universal route ranking.

478 Application to 196- and 900-dimensional coupled finite-element tangents
479 showed that the full coordinate–response–enclosure chain can qualify archived
480 application operators. The resulting implementation is a reproducible reference
481 benchmark for future sparse estimators, not a replacement for their independent
482 coverage, tightness, and cost validation.

483 Data availability

484 The non-confidential frozen operands, synthetic validation records, exact-
485 value tables, claim–evidence matrix, and provenance files supporting this study
486 are archived in Zenodo at <https://doi.org/10.5281/zenodo.21833850> [34]. This
487 concept DOI resolves to the latest published artifact version. The archive excludes
488 Abaqus executables, user-element source, solver state histories, and engineering
489 project data.

490 **Code availability**

491 The scripts for the analytical constructions, finite-precision error-enclosure
492 validation, and the two reference finite-element linearization applications are
493 available at [https://github.com/yqma7980/scale-explicit-matrix-perturbation-](https://github.com/yqma7980/scale-explicit-matrix-perturbation-observability)
494 observability and are preserved with the cited Zenodo release. Code is licensed
495 under the MIT License; data and documentation are licensed under CC BY 4.0.

496 **Reproducibility statement**

497 The reported calculations use immutable operand snapshots, explicit coor-
498 dinates and reference problems, two independent precision levels, independent-
499 process repeatability checks, and exact output records. The Supplementary
500 Information retains full numerical precision, error-enclosure components, and
501 execution provenance.

502 **CRedit authorship contribution statement**

503 **Bing Liang:** Conceptualization, Methodology, Formal analysis, Writing –
504 original draft, Writing – review and editing, Supervision. **Yangqi Ma:** Methodol-
505 ogy, Software, Validation, Formal analysis, Investigation, Data curation, Visual-
506 ization, Writing – original draft, Writing – review and editing. **Weiji Sun:** Concep-
507 tualization, Methodology, Resources, Supervision, Project administration, Fund-
508 ing acquisition, Writing – review and editing. **Shi He:** Methodology, Validation,
509 Writing – review and editing.

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518 **preparation process**

519 During preparation of this work, the authors used Codex (OpenAI desktop ver-
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